

Spreading 45 Points in the Plane

Limin Ge*

October 9, 2026

Abstract

We place 45 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 42.36644+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 46-point configuration gives 43.71615; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 45 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 42.36644\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 45$ is not on the page. Deleting a point from our 46-point configuration gives 43.71615; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n45.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
21183220525007025282506941
denominator:
499999999999128491953505
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

45 points, x and y per line. File: friedman-maxmin2d/coords/n45.txt.

```
0.740057331618 1.869222392277
-1.249038661075 1.660662732106
-0.073872700816 0.042334901953
3.839766880614 -0.012396278731
2.438253868050 -1.030119853978
1.342762702621 4.213444525462
1.500450771989 -1.377287469143
-0.422916638372 -0.894771457888
1.749727718838 5.126888270624
-0.842073644858 2.574106477268
2.517928662880 2.595116695309
3.331858695315 4.422004185633
3.512476659227 2.699396525395
1.930345682751 3.404280610386
0.152474351488 2.678386307353
1.523380666533 2.490836865224
1.670052924008 1.501651721336
-1.668376067278 3.455571759701
3.659148916701 1.710211381507
2.664600920355 1.605931551422
0.935797686404 3.300000780300
2.337310698968 4.317724355548
-1.072144573534 4.258384319311
3.252183900484 0.796767636345
4.498556541676 2.865668788355
-0.254490664729 1.764942562191
2.845218884267 -0.116676108817
4.101416217448 3.783426688279
-0.435108628641 3.487550222430
0.749612177046 5.088211927992
0.856122891574 -0.325235768987
-1.088382187106 -0.144174014196
-1.836621641205 2.469826647183
4.322327536877 0.910397624762
0.333092315401 0.955778647115
-0.288821238619 4.879998792258
-0.661455680946 0.851498817030
0.507078954018 -1.262342128829
1.263087907791 0.588207976175
2.257635904138 0.692487806260
4.645228799150 1.876483644468
1.850670887920 -0.220955938902
0.348214706274 4.109164695377
-1.656003677293 0.747218986944
2.924893679097 3.508560440471
```